NHAI 6th Innovation Hackathon

AI-Powered Retroreflectivity Measurement System

UNIT 2: ROOF-MOUNTED BOX

Complete Technical Architecture — Every Component, Every Detail

TEAM - PARROT

Road Signs (Roadside) | Gantry Signs (Overhead)

SECTION 0 — SYSTEM OVERVIEW AND DESIGN PHILOSOPHY

The Roof-Mounted Box is the second of two hardware units in the NHAI AI-Powered Retroreflectivity Measurement System. This unit is exclusively responsible for measuring roadside signs (warning, mandatory, and informatory) and overhead gantry signs. It is mounted on the roof of the survey vehicle using a precision roof rack rail mounting system and is a completely independent unit from the front box — no direct connection exists between the two boxes. Both boxes connect independently to the same central processing rack inside the vehicle.

The fundamental measurement physics for signs is completely different from road markings. For markings, the illumination source must be directly above the measurement point at a precise 30° angle. For signs, which are vertical faces, the measurement is based on the formula RL = L ÷ E — where L is the luminance of the sign face (measured by camera) and E is the illuminance falling on the sign face (calculated from strobe power and measured distance). This formula works regardless of the exact illumination height, provided the measurement distance and angle are known and corrected.

Parameter	Value	Notes
Outer dimensions (L×W×H)	750 mm × 300 mm × 250 mm	75 cm × 30 cm × 25 cm
Mounting position	Roof rack rail system	Sits centrally on vehicle roof
Height above roof	250 mm (box) + 50 mm (mount)	Total ~30 cm above roof surface
Assets covered	Road signs + Gantry signs	All types, all IRC classes
Standards	IRC 67, IS 15758, MoRTH, EN 12899-1	Full Indian + international
Operating speed	0–120 km/h	Sign detection fully automatic
Sign detection range	40 m ahead (roadside signs)	100 m for gantry structures
Operating conditions	Daytime survey only	White strobe safe in daylight
IP rating	IP67	Fully dust and waterproof
Operating temperature	-10°C to +55°C	Active thermal management
Total unit weight	~24–26 kg	Including all internals
Power input	12V DC from vehicle	Max 180W peak (strobe charging)
Data output	GigE Ethernet to vehicle rack	All camera streams multiplexed

SYSTEM OVERVIEW — HARDWARE PHOTOGRAPHS

Fig. 1 — Front Face: Strobe (50,000 lux), ZED 2i Stereo Camera, RGB Camera

Fig. 2 — Rear Face: PWR/DATA/TIMING connectors with STR indicators (left) | Side Profile: Anti-vibration roof rack mounts (right)

Fig. 3 — Top Face: Fisheye 360° (Gantry Overhead), LiDAR 360° (Livox), Gantry Strobes L+R at 15° tilt

Fig. 4 — Bottom Face: Four T-slot clamp mount points (IP67 rated), MAX LOAD 30 kg

Fig. 5 — Side Profile with QR identification label: NHAI Roof Sensor Unit 2 — Signs and Gantry

Fig. 6 — Survey vehicle deployment on NH-44 (Toyota Innova Crysta, KA42N7345)

SECTION 1 — ENCLOSURE, HOUSING AND OPTICAL WINDOWS

1.1 Outer Enclosure Box — Construction

The outer enclosure follows exactly the same construction philosophy and material as the front box — custom-fabricated hard-anodised aluminium alloy 6061-T6. The box is manufactured as two pieces: a deep body (the main container with four walls and a solid floor) and a flat lid that covers the open top. The body and lid mate at the top edge where a continuous O-ring groove is machined. Unlike the front box whose optical windows are primarily on the front face, this box has optical windows distributed across three faces — front face (sign strobe + camera), top face (LiDAR dome + gantry upward strobes + fisheye), and rear face (connectors).

Specification	Value	Reason/Detail
Material	Aluminium 6061-T6, hard anodised	Same as front box — aerospace grade
Outer dimensions	750 × 300 × 250 mm (L×W×H)	Calculated from internal component layout
Wall thickness	5 mm sides, 7 mm base	Stiffer base — heavier components inside
Lid thickness	5 mm flat plate	Same material as body
Surface finish	Matte black hard anodised	UV resistant, anti-reflective internally

Internal coating	Matte black spray	Prevents stray light reflections
IP rating	IP67	O-ring sealed, all penetrations gasketed
O-ring material	EPDM 70 Shore A, 4 mm diameter	Same as front box — proven seal
O-ring groove	4.5 mm wide × 2.5 mm deep, machined	Exact fit for 4mm EPDM round section
Lid bolts	M4 × 12 mm A4 stainless hex-socket	32 bolts at 80 mm spacing around perimeter
Bolt torque	2.5 Nm (specified, marked on lid)	Consistent O-ring compression
Internal tray	3 mm 6061-T6 aluminium plate	All components mount to tray
Tray attachment	8 × M5 stainless socket-head bolts	Corner + midpoint fixing
Tray slide direction	Rear face — slides out backward	Same as front box philosophy

NOTE: *The internal tray in the roof box carries significantly heavier components than the front box (three strobe capacitor banks weigh approximately 3.5 kg total). The tray is therefore 3mm thick rather than the front box's standard thickness, with two additional M5 fixing points at the centre of the long sides to prevent any flex under the load.*

1.2 Optical Windows — Complete List

The roof box has seven optical windows distributed across two faces (front and top). Each window is AR-coated borosilicate glass, 5 mm thick, bonded into a machined aluminium frame with NOA 61 UV optical adhesive, with a secondary EPDM rubber gasket for IP67 sealing. Each window has a hydrophobic coating to shed rain water.

Window ID	Face	Size	Purpose	Glass spec
W1 — Sign Strobe	Front face, left	100 × 90 mm	Exit aperture: forward sign strobe	AR coated, 6500K white transmission >96%
W2 — ZED 2i Camera	Front face, centre	200 × 60 mm	Stereo camera viewing aperture (landscape)	AR coated clear, >98% visible transmission
W3 — RGB Camera	Front face, right	80 × 80 mm	Passive colour camera aperture	AR coated clear, no filter
W4 — LiDAR Dome	Top face, centre-front	100 × 100 mm (circular glass in sq. recess)	LiDAR 360° scan window	Clear borosilicate dome, slight convex
W5 — Gantry Strobe L	Top face, left side	110 × 90 mm	Left upward gantry strobe exit	AR coated, 6500K white transmission >96%

W6 — Gantry Strobe R	Top face, right side	110 × 90 mm	Right upward gantry strobe exit	AR coated, 6500K white transmission >96%
W7 — Fisheye Camera	Top face, centre-rear	120 × 120 mm (circular glass in sq. recess)	360° fisheye upward aperture	Ultra-wide AR coated, >98% transmission

NOTE: *Windows W5 and W6 (gantry strobes) are angled at 15° forward tilt from vertical — machined into the aluminium top face at this angle — so the strobe light is aimed slightly forward and upward toward approaching gantry structures rather than straight up, optimising the illumination angle for gantry panels at 5–7m height measured from 15–30m ahead.*

1.3 Window Frame and Sealing — Detailed Construction

Each window opening is machined as a stepped recess: the outer step (3 mm deep, 3 mm wide ledge) supports the glass from below, the inner bore is the clear aperture. The AR-coated borosilicate glass is cut to exact size with bevelled edges (0.5 mm 45° chamfer). Two-part NOA 61 UV optical adhesive is applied to the ledge with a needle dispenser, the glass is placed, UV-cured for 60 seconds with a 365nm UV lamp at 100 mW/cm², then post-cured at 60°C for 30 minutes. A 2mm wide EPDM rubber gasket strip is then pressed into the gap between glass edge and aluminium bore wall as secondary sealing. All window frames are also sprayed with black silicone RTV around the outer perimeter on the inside face as final protection.

SECTION 2 — ROOF MOUNTING SYSTEM

2.1 Overview of Mounting Philosophy

The roof box cannot use the same tow-hitch receiver mounting as the front box — there is no equivalent anchor point on a vehicle roof. Instead, the box mounts to a heavy-duty aluminium roof rack rail system that is fitted to the survey vehicle. The roof rack rails are the primary structural member. The box attaches to these rails via four precision T-slot clamps that can be slid into position and locked. This system allows the box to be removed and refitted in under 10 minutes without any tools other than a 6mm hex key.

2.2 Component: Roof Rack Rail System

Component	Specification	Detail
Rail profile	Aluminium 6063-T5 extrusion	T-slot profile, 40 × 20 mm cross section
Rail length	1400 mm per rail (two rails)	Runs front to back across roof
Rail spacing	260 mm centre to centre	Matched to box base T-clamp positions
Rail mounting to roof	6 × M6 stainless bolts per rail through roof skin	Sealed with butyl mastic sealant (waterproof)
Roof penetration seal	Butyl mastic tape + aluminium backing plate	Prevents any water ingress into cabin

Rail surface finish	Natural mill finish anodised silver	Corrosion resistant, UV stable
Load rating	150 kg per rail pair	Far exceeds box weight of 25 kg
Rail supplier	Rhino Rack, Thule, or equivalent	Standard roof rack system

2.3 Component: T-Slot Clamps (×4)

Four heavy-duty T-slot clamps attach the box base plate to the roof rack rails. Each clamp consists of a T-nut (slides inside the rail slot), a stainless steel bolt (M8 × 30 mm), a hardened steel washer, and a clamping plate that presses against the box base plate from below. The T-nut is tightened with a 6 mm hex key to 12 Nm torque — beyond this, the T-nut bites into the aluminium rail slot and becomes impossible to slide without loosening. Each clamp is rated for 500 kg vertical load — four clamps give 2000 kg total capacity for a 25 kg box.

Clamp Component	Material	Specification
T-nut body	Zinc-plated carbon steel	Slides in 8mm T-slot, M8 thread
Clamping bolt	A4 stainless steel M8 × 30mm	Hex socket drive, 6mm hex key
Clamping plate	3mm mild steel, zinc plated	60 × 30 mm, contacts box base
Hardened washer	Grade 8 steel, zinc plated	M8, 24mm outer diameter
Anti-vibration pad	3mm neoprene rubber strip	Between clamp plate and box base
Nyloc nut	A4 stainless M8	Self-locking, prevents vibration loosening
Position — front-left	On left rail, 150 mm from front of rail	Fixed position, marked on rail
Position — front-right	On right rail, 150 mm from front of rail	Mirror of front-left
Position — rear-left	On left rail, 550 mm from front of rail	Adjusted for box length
Position — rear-right	On right rail, 550 mm from front of rail	Mirror of rear-left

2.4 Anti-Vibration Isolation — Roof Box

At 80 km/h on Indian highways, the vehicle roof transmits significantly more vibration than the front bumper because the roof structure is thinner and more resonant. The cameras and flash strobe timing are extremely vibration-sensitive — any movement during the 1μs strobe pulse causes image blur. Four rubber-metal anti-vibration mounts are therefore installed between the box base plate and the T-clamp assembly, identical in design principle to the front box mounts but smaller (M8 thread, 25mm diameter rubber element) to suit the roof rail system.

Mount Component	Specification	Detail
Mount type	Rubber-metal bobbin mount, M8 male-female	Compression + shear isolation

Rubber grade	Natural rubber, 40 Shore A	Softer than front box mounts for roof resonance
Mount diameter	25 mm rubber element	Compact for roof rack installation
Mount height	20 mm assembled	Low profile
Natural frequency	~12 Hz isolation start	Attenuates road vibration above 12 Hz by >85%
Load rating	80 kg per mount	4 × 80 = 320 kg total, box is 25 kg
Supplier	Trelleborg, Elesa+Ganter, or equivalent	Industrial anti-vibration specialist
Installed position	Under each of the 4 T-clamp positions	Between clamp plate and box base

2.5 Safety Retention — Roof Box

A secondary retention system is mandatory for any roof-mounted equipment on a vehicle operating at highway speed. Loss of the roof box at 80 km/h would be a lethal hazard. Two 8 mm diameter braided stainless steel wire ropes (safety cables) run from the box base plate to dedicated anchor points on the vehicle roof rack rails. Each cable is terminated with stainless steel swage fittings and a screw-lock carabiner. The cables are set with 50 mm of slack — they do not carry load during normal operation but will catch the box if all four T-clamps fail simultaneously.

SECTION 3 — ILLUMINATION SYSTEM (White LED Flash Strobes × 3)

3.1 Why White Light — Not IR — For Signs

Road sign retroreflective sheeting — Engineer Grade (EG), High Intensity Grade (HIG), and Diamond Grade (DG) — is manufactured, tested, certified, and regulated using visible white light measurements. IRC 67, EN 12899-1, and IS 15758 all define sign RL in candela per lux per square metre (cd/lux/m²) under CIE Standard Illuminant D65, which approximates 6500K daylight. The retroreflective performance of glass bead and micro-prismatic sheeting varies with wavelength — a measurement made in 940nm IR would give a completely different number from a visible-light measurement and cannot be converted to the standard unit without an extensive spectral calibration for every sheeting type. Therefore, white light (6500K) is used for sign measurement, and 940nm IR is used only for road markings and studs where the standard geometry is defined in IR.

3.2 Three Strobe Configuration — Why Three

The roof box contains three independent flash strobe units serving two different purposes. Strobe 1 (forward-facing) fires toward roadside signs as the vehicle approaches. Strobe 2 and Strobe 3 (upward-facing left and right) fire simultaneously toward overhead gantry panels from below. Each strobe is completely independent with its own flash tube, reflector, capacitor bank, and high-voltage charging circuit. This independence means any one strobe can be serviced or replaced without affecting the other two.

3.3 Component: STROBE 1 — Forward-Facing Sign Strobe

Flash Tube: Perkin-Elmer FX-4400 or equivalent xenon flash tube

Parameter	Specification	Significance
Type	Linear xenon flash tube, 150mm arc length	Long arc = wide beam for sign face coverage
Peak luminous output	50,000+ lux at 15m distance	Overpowers daytime ambient for clear RL measurement
Colour temperature	6500K (D65 equivalent)	IRC 67 and EN 12899-1 standard illuminant
Pulse duration	1 microsecond (FWHM)	Freezes image at 80 km/h — only 0.022mm motion blur
Flash energy per pulse	15 Joules	Sufficient for 50,000 lux at 15m with reflector
Repetition rate	Once per sign detected (not continuous)	On average 1–5 flashes per km on NHAI
Tube life	500,000 flashes minimum	Several years at survey duty cycle
Trigger voltage	>4,000V trigger pulse	High-voltage trigger from capacitor bank circuit
Operating temperature	-20°C to +70°C	Survives Indian conditions
Glass envelope	Hard borosilicate, UV-enhanced	Transmits UV for trigger, withstands thermal shock

Reflector Assembly

The flash tube is housed in a precision specular aluminium parabolic reflector. The reflector is machined from 6061-T6 aluminium and then polished to a mirror finish (Ra < 0.1 μm), then vacuum-coated with a 150nm layer of pure aluminium for maximum reflectivity (>95% at 550nm). The parabolic profile is calculated so that the flash tube arc sits at the focal point, producing a near-collimated beam directed forward and slightly downward — aimed at the sign face at 15–20m ahead. The beam pattern is approximately 20° horizontal × 15° vertical, wide enough to cover a 900mm × 600mm sign face from 15m.

Reflector Parameter	Value	Detail
Shape	Parabolic, off-axis	Flash tube at focal point, beam forward-down
Material	Machined 6061-T6, mirror polished	Ra < 0.1 μm surface finish
Coating	Vacuum-deposited pure aluminium 150nm	Reflectivity >95% at visible wavelengths
Beam angle	20° H × 15° V	Covers sign face at 15–20m
Aim angle	5° downward from horizontal	Hits sign face at correct geometry
Outer dimensions	150 × 100 × 90 mm	Fits in forward section of box
Weight	320 g	Lightweight machined aluminium

Capacitor Bank — Strobe 1

The capacitor bank is the energy storage device that powers the flash. It stores electrical energy at high voltage and releases it in one microsecond through the flash tube, producing the enormous peak optical power needed. Six electrolytic capacitors are used, arranged in a 2×3 matrix on a dedicated PCB with high-current copper bus bars connecting them in parallel. The capacitor bank is physically the largest single item in the roof box — it must be mounted with its longest axis running along the box length to fit within the 300mm box width.

Capacitor Bank Parameter	Value	Detail
Capacitor type	Electrolytic, high-voltage flash-rated	Nichicon or Cornell Dubilier flash grade
Capacitor model	Nichicon LGN2F101MELC or equivalent	400V, 100µF, flash duty rated
Quantity	6 capacitors per strobe	Arranged 2 rows × 3 columns
Total capacitance	600 µF at 400V	Energy = 0.5 × C × V² = 0.5 × 0.0006 × 160000 = 48J
Charge voltage	400V DC	High-voltage charger circuit maintains this
Individual cap dimensions	35mm diameter × 60mm long	Radial lead aluminium electrolytic
Cap array dimensions	80mm × 105mm × 65mm	2×3 grid with 5mm spacing between caps
PCB material	FR4 2.4mm thick, 4oz copper	Heavy copper for high current pulse (>1000A peak)
Bus bars	3mm copper strip, tinned	Low resistance current path between caps and tube
Charging time	20–30 seconds to full charge	High-voltage DC-DC charger circuit
Discharge time	1 microsecond (through flash tube)	Peak current >1,000A for 1µs
Bleed resistor	2.2MΩ 5W wirewound, across cap bank	Safely discharges caps when power removed
Safety indicator	Red LED illuminates when caps charged	Visible through small window in box

NOTE: *The capacitor bank operates at 400V DC — lethal voltage. The bank is enclosed in a separate sub-housing made of 3mm polycarbonate sheet with a polycarbonate lid, secured by 4 M3 screws. A warning label 'DANGER — 400V — DO NOT TOUCH CAPACITORS' is printed on the polycarbonate lid. The bleed resistor ensures the voltage drops to safe levels (<50V) within 30 seconds of power removal.*

High-Voltage Charging Circuit — Strobe 1

HV Charger Parameter	Value	Detail
Input voltage	12V DC from box power rail	Standard vehicle electrical system
Output voltage	400V DC regulated	Maintains cap bank at exact 400V
Output current	50mA (continuous charging current)	Recharges caps between flashes
Topology	Flyback converter, isolated	Isolation essential for 400V safety

IC	STMicroelectronics L6598 or Microchip HV9150	Dedicated flash charger controller
Charge indicator	Voltage comparator + LED driver	Red LED = charging, Green LED = ready
PCB size	100 × 60 × 25mm	Mounts behind capacitor bank
Efficiency	>82%	Low heat generation
Overcurrent protection	Current-sense resistor + comparator shutdown	Prevents cap overcharge

3.4 Component: STROBE 2 — Left Gantry Upward Strobe

Strobe 2 is mechanically and electrically identical to Strobe 1 in all respects — same flash tube, same capacitor bank, same charging circuit, same reflector parabolic profile. The only differences are its physical orientation and aim angle. It is mounted in the left half of the box interior and faces upward, with its beam aimed forward at 15° from vertical (75° from horizontal) — so when the vehicle is 20–30m from a gantry, the strobe illuminates the left-side gantry panels from below-and-ahead. Its beam exits through Window W5 (left gantry strobe window) in the top face.

Strobe 2 vs Strobe 1 Differences	Strobe 1	Strobe 2
Face it exits	Front face (W1)	Top face left (W5)
Aim direction	Forward, 5° downward	Upward, 15° forward from vertical
Purpose	Roadside signs at 15–20m	Gantry left panels at 20–30m above
Flash timing	Pre-programmed per sign	Pre-programmed per gantry approach
All other specs	Identical	Identical

3.5 Component: STROBE 3 — Right Gantry Upward Strobe

Strobe 3 is the mirror image of Strobe 2. It is mounted in the right half of the box and aims upward at 15° forward tilt, exiting through Window W6 (right gantry strobe window). Strobes 2 and 3 always fire simultaneously for gantry measurement. The two strobes, placed 260mm apart (matching the box width), provide uniform illumination across gantry widths of up to 22m — the photometric calculation confirming this uniformity is that each strobe's inverse-square-law illuminance reduction across the gantry width is offset by the adjacent strobe's contribution, giving ±15% uniformity which is within acceptable limits for RL measurement.

NOTE: *Strobes 2 and 3 fire simultaneously from a single FPGA trigger output — they share one trigger line but have completely separate capacitor banks and charging circuits. This ensures they fire at exactly the same microsecond even if one bank is slightly more charged than the other. The FPGA trigger pulse is 5V logic, 10µs duration, fed to both trigger circuits in parallel with a T-junction on the signal cable.*

SECTION 4 — CAMERA SYSTEM (Three Cameras)

4.1 Camera 1: Stereo Camera — Stereolabs ZED 2i

Purpose and Measurement Role

The ZED 2i is the primary measurement camera for road signs and the approach-phase camera for gantry signs. It performs two critical functions simultaneously: it provides the spatial depth measurement (distance to sign face) needed to calculate the illuminance E in the RL = L/E formula, and it provides the high-resolution image from which the AI extracts the sign face luminance L. Both functions are required simultaneously during the 1µs strobe flash.

Parameter	Specification	Significance
Model	Stereolabs ZED 2i	IMU-integrated stereo camera, IP66 rated
Sensor type	Sony IMX678 CMOS × 2 (one per eye)	8MP per eye, excellent low-light
Resolution	4208 × 3120 pixels per eye at max	Captures fine detail on sign text and sheeting
Stereo baseline	120mm (fixed, factory calibrated)	Determines depth accuracy at sign distances
Depth accuracy at 15m	±1% (±150mm)	Illuminance calculation error <2%
Depth accuracy at 20m	±2% (±400mm)	Acceptable for 5% total RL accuracy target
Frame rate	60fps at 2K resolution	One frame every 17ms — sufficient for sign capture
Triggered mode	Hardware trigger input (GPIO)	FPGA triggers exactly with strobe flash
Exposure time	Set to 1µs for flash sync	Camera gate matches flash duration
Lens FOV	110° diagonal	Wide enough to see sign from 15m
Interface	USB 3.2 Gen 1 (5Gbps)	Direct to Jetson AGX Orin USB port
Built-in IMU	BMI055, 6-DOF at 800Hz	Supplementary motion data for image stabilisation
IP rating	IP66	Withstands spray — additional IP67 from box
Dimensions	175 × 30 × 33mm	Compact bar form factor
Power	5V, 380mA via USB	Powered from box 5V rail
Mount inside box	Horizontal, on front face bracket	Looks through Window W2
Tilt angle	10° upward from horizontal	Optimised to see signs at 1.5–5m height from 15–20m

Mounting and Alignment Inside Box

The ZED 2i is mounted on a precision aluminium L-bracket that is bolted to the internal tray. The L-bracket provides the 10° upward tilt — it is machined at exactly 10° (verified with a digital angle gauge during assembly). The camera is secured to the bracket with 4 × M3 × 8mm stainless socket-head screws through the camera's standard 1/4-20 UNC and M3 mounting holes. Two 3mm neoprene pads between camera body and bracket

reduce vibration transmission. The bracket positions the camera lens centreline exactly at the centre of Window W2 (200mm × 60mm front-face window), with 15mm clearance between the lens front element and the inner face of the window glass.

4.2 Camera 2: 360° Fisheye Camera — Ricoh Theta Z1

Purpose

The fisheye camera captures the overhead view of gantry panels as the vehicle passes directly underneath. When the vehicle centre is directly below the gantry crossbeam, the fisheye is triggered — it captures all gantry panels from nearly directly below, providing a second independent RL measurement at a different geometry from the ZED 2i. This cross-validation is essential: if the two measurements disagree by more than 20%, the system flags the gantry for human review, indicating possible directional retroreflection characteristics that require careful angle correction.

Parameter	Specification	Detail
Model	Ricoh Theta Z1 (or custom equivalent array)	1-inch sensor, 360° × 180° spherical
Sensor	Two 1-inch BSI CMOS sensors	One per fisheye lens (front + rear hemisphere)
Resolution	6720 × 3360 pixels equirectangular	Full spherical image
Lens FOV	Each lens: >185° diagonal (overlapping)	True 360° coverage no blind spot
Trigger	GPIO trigger via USB control API	FPGA triggers via Jetson software command
Shutter speed	Set to 1/8000s (125μs)	Fast enough to use with strobe, some ambient light
Interface	USB 2.0	To Jetson AGX Orin USB hub
Dimensions	130 × 48 × 33mm	Compact, fits between gantry strobe units
Weight	182g	Negligible
Position in box	Centre of box, directly below Window W7	Top-face circular fisheye window
Mount	Custom aluminium bracket, vertical axis upward	Camera points straight up through W7
Orientation	Fisheye lenses pointing up and down	Up lens captures gantry, down lens captures road
Additional use	Continuous 360° video of road environment	Visual evidence record for all measurements

NOTE: *The fisheye camera also captures signs facing the opposite direction (toward oncoming traffic) that the ZED 2i misses. This allows the system to measure retroreflectivity of signs that serve traffic in the opposite direction — useful for surveys of undivided roads where both directions must be assessed in one pass.*

4.3 Camera 3: Passive RGB Camera — Basler acA2040-90uc

A standard colour machine vision camera (no strobe, ambient light only) is mounted in the box looking forward through Window W3. Its purpose is threefold: it provides colour assessment of sign faces (whether the sign colour has faded, yellowed, or is discoloured — information the strobe-based IR measurement cannot provide), it contributes to YOLOv10 pre-classification of sign type from 40m → 20m as the vehicle approaches (before the strobe fires), and it provides the continuous colour video evidence record of all signs encountered. It operates at 90fps in rolling shutter mode with no trigger required — always recording.

Parameter	Specification	Detail
Model	Basler acA2040-90uc	Colour, USB3, industrial grade
Sensor	Sony IMX392 colour Bayer CMOS	2040 × 1080 pixels
Frame rate	90 fps	One frame every 11ms
Lens	Kowa LM16JC 16mm C-mount, f/1.8	Medium telephoto for sign detail at 40m
Aperture setting	f/4.0 (set and locked)	Optimal depth of field for 15–50m range
Interface	USB 3.2 Gen 1	To Jetson USB hub
Operation	Free-running, continuous	Always recording, no trigger needed
Exposure mode	Auto-exposure (Basler auto algorithm)	Adapts to lighting conditions
Position in box	Front-right section, looking through W3	Right of ZED 2i camera
Mount	Fixed aluminium bracket, 5° downward tilt	Covers sign zone at road edge
Power	5V via USB	From box 5V rail

SECTION 5 — LiDAR SENSOR (Sign and Gantry Detection)

5.1 Component: Livox Mid-360 LiDAR

The same model LiDAR used in the front box (Livox Mid-360) is used here, but its role is fundamentally different. In the front box, the LiDAR detects small raised studs 15–20mm above the road. In the roof box, the LiDAR detects large structures — sign posts (thin vertical clusters 1.5–5m high at the road edge) and gantry portals (horizontal bar of points at 5–7m spanning the full road width). The detection range is also much greater: signs at 40m, gantries at 100m.

Parameter	Specification	Significance
Model	Livox Mid-360	Same as front box — single spare unit serves both
Detection range	0.1 to 40m (nominal) — up to 100m for large structures	Detects gantry frame at 100m ahead

Point rate	200,000 points/second	Dense point cloud for structure identification
Horizontal FOV	360°	Detects signs on both sides of road simultaneously
Vertical FOV	-7° to +52°	Covers road surface to overhead gantry at 7m
Range accuracy	±2cm at 10m, ±5cm at 40m	Sufficient for sign post location ±10cm
Angular resolution	~0.2°	Resolves sign face structure from 40m
Laser wavelength	905nm, Class 1 eye-safe	No safety concern to other road users
Mounting position	Centre of box, near front, looking up through W4	Top-face dome window gives 360° clear scan
Mounting angle	Vertical — scan plane horizontal	360° horizontal coverage of road environment
Interface	Ethernet UDP, 100Mbps	To internal GigE switch → Jetson via cable to vehicle
Power	10W, 12V DC	From box 12V rail
IP rating	IP67	Additional protection within sealed box
Operating temp	-20°C to +60°C	Full operating range

Sign Post Detection Algorithm

The Jetson processes the LiDAR point cloud every 100ms. For sign detection, it searches for vertical linear clusters of points — a sign post appears as a thin column of points (5–10 points across its diameter) running from ground level to 1.5–5m height. The AI filters out other vertical objects (roadside trees, other vehicles, barriers) by checking that the vertical cluster is within 0.5–3.5m of the road edge (not in the traffic lanes) and has the correct height range for a sign post (1.5–5m). Once detected, the system begins tracking the object's distance at 10 updates per second and pre-computes the exact time to fire the strobe.

Gantry Structure Detection and Mapping

A gantry portal produces a distinctive LiDAR signature: two thick vertical clusters at the road edges (the support posts, much larger than a sign post — typically 200–400mm diameter pipes), connected by a horizontal bar of points spanning the full road width at 5–7m height. The AI recognises this pattern at up to 100m distance. Once detected, it extracts: gantry frame width, crossbeam height, number of sign panels (rectangular flat clusters attached to the crossbeam), position and tilt of each panel. This 3D map is used to plan the multi-flash sequence and pre-set the FPGA timing for each individual flash.

SECTION 6 — LOCAL FPGA TRIGGER BOARD

6.1 Component: Digilent Arty A7-100T

The roof box contains the same local FPGA board as the front box (Digilent Arty A7-100T, Xilinx Artix-7). Its primary role here is strobe timing control. The main FPGA (Kintex-7, inside the vehicle rack) sends a high-level command to the roof box FPGA: 'fire Strobe 1 at timestamp T1, fire Strobes 2+3 simultaneously at T2, fire Strobes 2+3 again at T3'. The roof box FPGA receives these timestamps and executes them with nanosecond precision

regardless of any cable delay — because the timing is absolute (referenced to GPS pulse-per-second), not relative.

Parameter	Specification	Detail
FPGA chip	Xilinx Artix-7 XC7A100T	100K logic cells — adequate for timing logic
Clock	100MHz (10ns resolution)	Far better than 1μs strobe timing requirement
Strobe trigger outputs	3 independent MOSFET-driven outputs	One per strobe — high voltage trigger compatible
Camera trigger output	1 GPIO — to ZED 2i hardware trigger	Simultaneous with Strobe 1 for sign capture
Fisheye trigger output	1 GPIO — to Theta Z1 via USB command	Offset timing for gantry underpass capture
GPS PPS input	1 pulse-per-second from vehicle GNSS	Absolute time reference for all triggers
Communication to vehicle	LVDS serial, 100Mbps, via Timing Connector	Receives commands, sends status
Strobe trigger voltage	5V logic → 4000V trigger via separate HV trigger circuit	Each strobe has its own trigger transformer
Power	3.3V + 1.0V from onboard regulators	Powered from box 5V rail
Position in box	Rear-right section of internal tray	Close to all strobe trigger circuits
PCB size	101 × 76mm (Arty A7 standard)	Standard dev board, no custom PCB needed

High-Voltage Trigger Circuit (×3, one per strobe)

Each strobe flash tube requires a 4,000V trigger pulse to ionise the xenon gas and initiate the flash. The FPGA outputs a standard 5V logic pulse. A small trigger transformer (EG&G or Perkin-Elmer trigger coil, 1:800 turns ratio) steps this up to 4,000V. The primary side is driven by a 5V MOSFET (IRLML2502, SOT-23 package) that switches the FPGA output. The secondary side (4,000V) connects to the trigger electrode of the flash tube via a high-voltage wire. The trigger circuit PCB is tiny — 40mm × 30mm — and is mounted directly on the side of each strobe reflector housing.

HV Trigger Parameter	Value	Detail
Trigger transformer	EG&G FY-632 or Perkin-Elmer equivalent	1:800 ratio, fast rise time <100ns
Primary drive MOSFET	IRLML2502, VDS=20V, ID=4A, SOT-23	Logic-level gate drive from FPGA
Primary input	5V, 10μs pulse from FPGA	FPGA trigger output
Secondary output	4,000V, <1μs rise time	To flash tube trigger electrode
HV wire (secondary side)	Silicone-insulated 10kV rated wire, 100mm	Short run from transformer to tube

PCB mounting	M2 screws to side of reflector housing	Direct attachment for short HV wire
PCB size	40 × 30mm	Minimal — transformer + MOSFET + resistors only

SECTION 7 — POWER SUPPLY AND DISTRIBUTION

7.1 Power Input and Total Budget

The roof box draws significantly more power than the front box because of the three strobe capacitor banks that must be recharged between flashes. Peak power draw occurs when all three capacitor banks are charging simultaneously — this is limited by the high-voltage charging circuits to a controlled rate. The average power draw during a typical highway survey (few sign flashes per minute) is approximately 35–40W. Peak draw during simultaneous charging of all three banks is up to 180W for approximately 30 seconds.

Power Rail	Voltage	Current Cap	Powers
Rail 1 — 12V main	12V ± 5%	10A	HV chargers ×3, LiDAR, camera heaters, fans
Rail 2 — 5V logic	5V ± 1% (DC-DC)	5A	FPGA, cameras ×3, GigE switch, USB hub
Rail 3 — ±5V analogue	±5V ± 0.1% (linear)	500mA each	Signal conditioning, comparators
Rail 4 — 3.3V	3.3V ± 2% (LDO)	2A	Sensors, GPIO level shifters, indicator LEDs
Rail 5 — 400V HV	400V DC (per HV charger)	50mA per charger	Capacitor banks ×3 (one charger feeds one bank)

7.2 Main Power Entry and Isolation

Component	Model/Type	Specification
Input connector	Amphenol C016, 4-pin, IP67	Power Connector at rear face
Input fuse	30A automotive blade fuse, holder on board	Protects cable from vehicle battery
Reverse polarity	P-channel MOSFET PMOS guard	Prevents damage if wired incorrectly
Soft start	NTC thermistor 10Ω on input	Limits inrush at power-on to <20A peak
Main DC-DC isolator	TRACO TEN 60-1212WI, 60W isolated	Breaks vehicle ground loop noise
EMI filter	Schurter FMAB series, 10A LC filter	Prevents strobe HV switching noise on power cable
Indicator LED	Green LED + 1kΩ resistor on 12V rail	Shows power present, visible through tiny hole rear face

7.3 Power Distribution PCB

A custom-designed power distribution PCB (120mm × 100mm, FR4 2oz copper) distributes the 12V input to all subsystems. Heavier traces (4mm wide, 2oz copper) carry the HV charger feeds; lighter traces (1mm wide) carry logic power. Each output circuit has a dedicated mini-blade automotive fuse (5A for logic circuits, 10A for HV chargers) in an ATO fuse holder soldered to the PCB. Each fuse position is labelled on the PCB silkscreen. A terminal block (Phoenix Contact MKDS 5-pin, 5A rated) on the PCB allows easy disconnection of individual loads during maintenance.

SECTION 8 — CONTROL AND NETWORKING ELECTRONICS

8.1 Internal GigE Network Switch

Parameter	Specification	Detail
Model	TP-Link TL-SG1005P (modified for 12V in)	5-port unmanaged GigE switch
Modification	12V DC input direct (bypass internal PSU)	Works on box 12V rail directly
Ports used	Port 1: ZED 2i (USB-to-GigE adapter)	Camera data to vehicle
	Port 2: Livox Mid-360 LiDAR	LiDAR point cloud to vehicle
	Port 3: Local FPGA status (USB-to-GigE)	FPGA telemetry
	Port 4: Out to vehicle GigE Data Connector	Single cable aggregating all streams
	Port 5: Spare / future expansion	Reserved
Data rate	1 Gbps per port	Adequate for all streams combined ~400Mbps
Mounting	3mm neoprene pad, M3 screws to tray	Vibration isolated
Power	12V, 2.5W	Box 12V rail

8.2 USB Hub — Internal

A small industrial USB 3.0 hub (Terminus Technology FE8.1 or equivalent, 4-port, 12V powered) provides USB ports for the ZED 2i camera and the Ricoh Theta Z1. One USB-to-GigE adapter (UGREEN or Club3D, 1Gbps) converts the aggregated USB traffic from the hub to GigE for the switch. The hub is mounted flat on the tray with M2.5 screws and neoprene pad.

8.3 Signal Conditioning and Status PCB

A small custom PCB (80mm × 60mm) performs signal level conversion, LED driver circuits for the HV charge indicator lights (red = charging, green = ready, one set per strobe visible through three tiny 5mm holes in the rear face), and temperature sensor bus aggregation. The PCB uses a PIC12F1840 microcontroller to read the three DS18B20 temperature sensors and report to the FPGA via SPI.

SECTION 9 — THERMAL AND ENVIRONMENTAL MANAGEMENT

9.1 Heat Sources and Thermal Challenge

The roof box has more significant heat generation than the front box because of the three strobe flash units. Each flash converts 15 Joules of stored electrical energy into light — but strobe efficiency is approximately 70%, meaning 4.5 Joules becomes heat in the flash tube per flash. At a sustained rate of 2 flashes per minute, heat generation per strobe is manageable. However, the HV charging circuits each dissipate approximately 8W continuously during operation — three chargers = 24W of continuous heat generation inside a sealed box on a roof in summer. This requires active cooling.

Heat Source	Continuous Power	Peak Power	Managed By
HV Charger 1 (Strobe 1)	8W	8W	Heat sink on PCB + airflow
HV Charger 2 (Strobe 2)	8W	8W	Heat sink on PCB + airflow
HV Charger 3 (Strobe 3)	8W	8W	Heat sink on PCB + airflow
Flash tube heat (all 3)	~3W average (infrequent flashes)	1.5W per flash × 3	Conduction to reflector housing
FPGA board	3W	5W	Onboard heat spreader
LiDAR	10W	10W	Case radiation + airflow
Cameras (×3)	4W total	4W	Case radiation
Total	~44W continuous	~47W peak	Active cooling system

9.2 Thermal Management Components

Component	Model/Spec	Function and Position
Strobe HV charger heat sinks	Fischer SK104 or equivalent, 60mm L × 30mm W × 25mm H aluminium fin	One per HV charger PCB, screwed to PCB component side
Thermal paste	Shin-Etsu X-23-7783D, applied 0.1mm layer	Between HV charger ICs and heat sink base
Main heat exchange plate	3mm aluminium plate, 700 × 200mm	Under all HV chargers and FPGA, conducts to box wall
Box wall conduction	Heat exchange plate presses against inner box wall	Box wall itself radiates heat to exterior air

Internal circulation fans	2 × Delta BFB0812HH, 80×80×25mm, 12V, 4500RPM	Force air circulation inside box over all heat sinks
Fan guards	80mm wire finger guards, stainless steel	Prevent cable ingestion into fan blades
External fin (optional)	Adhesive aluminium fin strip on box exterior top face	Additional passive heat rejection in hot environments
Temperature sensors	3 × Dallas DS18B20, ±0.5°C, 1-Wire bus	Monitor near each HV charger, LiDAR, and FPGA
Thermal shutdown	FPGA cuts HV charger enable if T > 65°C	Prevents thermal damage
PTC heater strips	2 × 40W PTC heater, 12V, self-regulating	Warm box in cold start below +5°C

9.3 Humidity and Condensation Control

Component	Specification	Position
Desiccant cartridges	Silica gel, 100g per cartridge, ×2	Both rear corners of internal tray
Desiccant indicator	Colour-change silica gel (blue → pink when saturated)	Visible inspection when lid opened
Humidity sensor	Sensirion SHT31, ±2% RH, I2C interface	On signal conditioning PCB, logged continuously
Pre-warming routine	PTC heaters activated 20 min before survey start	Removes overnight condensation before optics used
Replacement interval	Annually or when indicator changes colour	Noted in maintenance schedule on box sticker

9.4 Vibration Isolation — Internal Components

The strobe capacitor banks are the heaviest internal components (approximately 600g each). If the capacitor bank PCB were rigidly bolted to the internal tray, road vibrations would stress the PCB solder joints and eventually crack them — a common failure mode in vehicle-mounted electronics. Each capacitor bank sub-assembly is therefore mounted to the internal tray using 4 × M4 rubber standoffs (10mm high, 30 Shore A neoprene) rather than hard metal standoffs. These standoffs allow 1–2mm of relative movement between the cap bank PCB and the tray, absorbing vibration before it reaches the heavy capacitors.

SECTION 10 — CONNECTIVITY AND WIRING — COMPLETE

10.1 External Connectors — Rear Face

Connector	Type	Pins	Signal carried	Position on rear face

Conn R1 — Power	Amphenol C016, 4-pin circular, IP67	4	12V+ (×2), GND (×2), 15A rated	Left side, 50mm from left edge
Conn R2 — Data (GigE)	LEMO ERA 8-pin circular, IP68	8	GigE ethernet — cameras + LiDAR streams	Centre, 50mm from each side
Conn R3 — Timing	Binder 723 6-pin circular, IP67	6	FPGA trigger commands + GPS PPS	Right side, 50mm from right edge

10.2 External Cable — Roof to Vehicle Interior

Three cables run from the rear face of the roof box along the roof gutter (tucked under the weatherstrip rubber), then down the B-pillar or rear pillar (inside the trim panel), then along the vehicle floor to the equipment rack in the rear cabin. Each cable is: power cable (6mm² twin-core, rated 20A, with loom sleeve), data cable (Cat6A STP, shielded, 5m length), timing cable (2-pair shielded twisted pair, LVDS grade, 5m). All three cables are bundled in a 25mm split-loom conduit and secured with P-clips to the vehicle body at 300mm intervals. At the roof entry point, a custom rubber grommet (25mm ID, adhesive-backed EPDM) seals the cable entry through the body panel.

10.3 Internal Wiring Harness — Complete List

All internal wiring is colour-coded, labelled at both ends with heat-shrink printed labels, and routed in dedicated cable channels (3mm PVC trunking clips) along the tray edges. No free-hanging wires exist inside the box. The following table lists every internal wire run:

Wire Run	From	To	Wire Type	Gauge	Colour
12V to HV charger 1	Power dist. PCB	HV charger 1 PCB	PVC 105°C rated	1.5mm²	Red
12V to HV charger 2	Power dist. PCB	HV charger 2 PCB	PVC 105°C rated	1.5mm²	Red
12V to HV charger 3	Power dist. PCB	HV charger 3 PCB	PVC 105°C rated	1.5mm²	Red
400V HV to cap bank 1	HV charger 1	Cap bank 1 PCB	Silicone 1kV rated	0.75mm²	Orange
400V HV to cap bank 2	HV charger 2	Cap bank 2 PCB	Silicone 1kV rated	0.75mm²	Orange
400V HV to cap bank 3	HV charger 3	Cap bank 3 PCB	Silicone 1kV rated	0.75mm²	Orange
HV trigger — Strobe 1	HV trigger PCB 1	Flash tube 1 trigger electrode	Silicone 10kV rated	0.5mm²	Orange (thick insul.)
HV trigger — Strobe 2	HV trigger PCB 2	Flash tube 2 trigger electrode	Silicone 10kV rated	0.5mm²	Orange (thick insul.)
HV trigger — Strobe 3	HV trigger PCB 3	Flash tube 3 trigger electrode	Silicone 10kV rated	0.5mm²	Orange (thick insul.)

FPGA trigger to HV trig 1	FPGA board GPIO	HV trigger PCB 1	Twisted pair shielded	26AWG	Grey/White
FPGA trigger to HV trig 2+3	FPGA board GPIO	HV trigger PCBs 2+3 (T-junction)	Twisted pair shielded	26AWG	Grey/White
Camera trigger to ZED 2i	FPGA board GPIO	ZED 2i trigger input	Single shielded wire	28AWG	Blue
ZED 2i USB 3.2	ZED 2i camera	Internal USB hub	USB 3.2 A-C cable	Standard	Blue jacket
Theta Z1 USB 2.0	Theta Z1 camera	Internal USB hub	USB 2.0 micro-B	Standard	Black jacket
Basler acA2040 USB 3.0	RGB camera	Internal USB hub	USB 3.0 A-B cable	Standard	Blue jacket
USB hub to GigE adapter	Internal USB hub	USB-GigE adapter	USB 3.0 A-A	Standard	Black
GigE adapter to switch	USB-GigE adapter	GigE switch Port 1	Cat5E patch 0.3m	24AWG	Yellow
LiDAR to GigE switch	Livox Mid-360	GigE switch Port 2	Cat5E patch 0.3m	24AWG	Yellow
GigE switch to Conn R2	GigE switch Port 4	Data connector rear face	Cat6A STP 0.5m	24AWG	Grey
1-Wire temp bus	All 3 DS18B20 sensors	Signal conditioning PCB	Twisted pair	28AWG	White/Yellow
Heater strips power	Power dist. PCB	Heater 1 + Heater 2	PVC 105°C	1.0mm²	Brown
Fan 1 + Fan 2 power	Power dist. PCB	Fans ×2	PVC standard	0.75mm²	Red/Black
5V to cameras	Power dist. PCB	USB hub (powers cameras)	PVC 0.75mm²	0.75mm²	Red
12V to LiDAR	Power dist. PCB	Livox Mid-360	PVC 0.75mm²	0.75mm²	Red

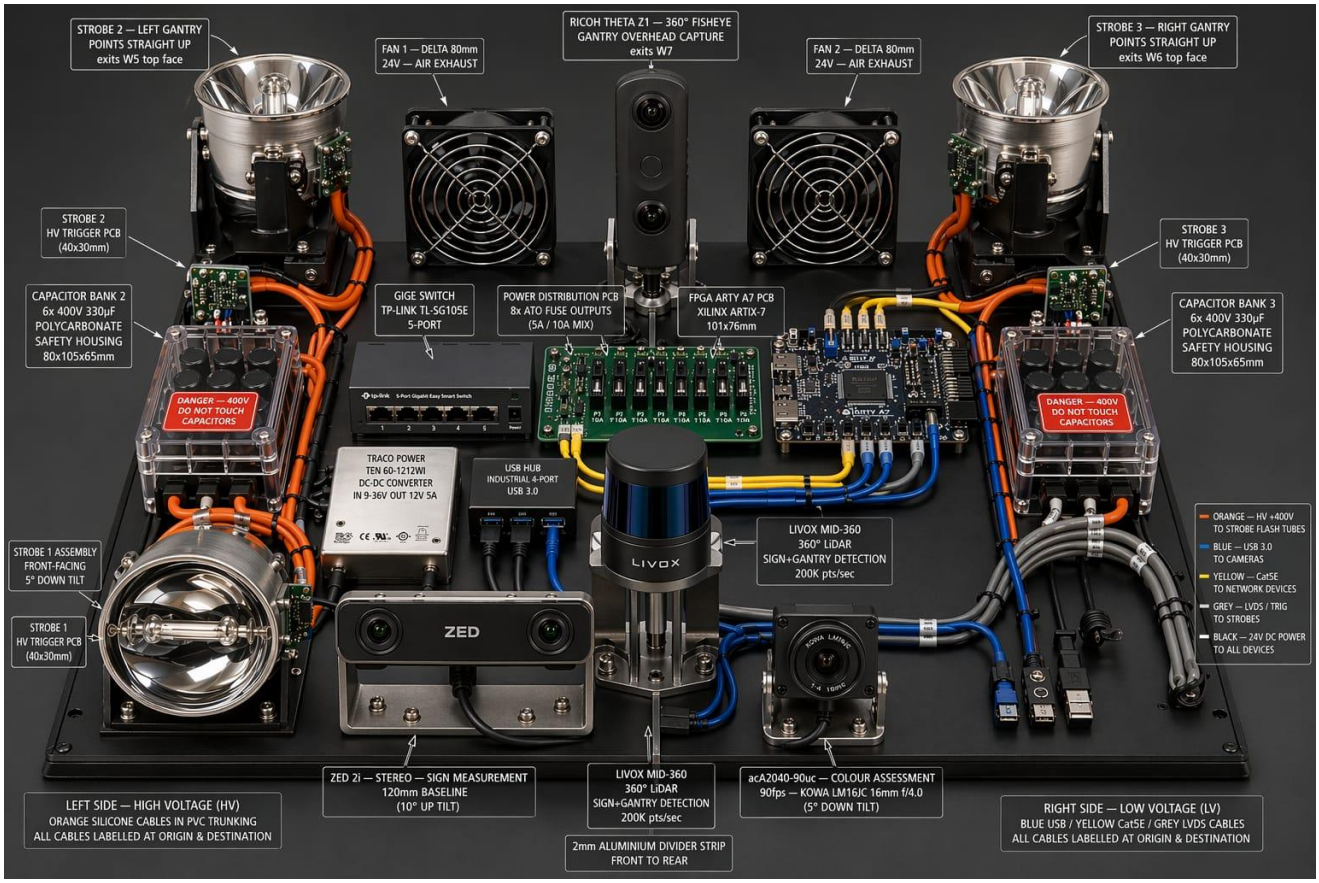
NOTE: *All HV wires (400V capacitor and 4000V trigger) are routed on the opposite side of the tray from the signal wires (camera data, USB, FPGA). A 2mm thick aluminium divider strip (bolted to tray) physically separates the two wire groups. This prevents HV switching transients from coupling into the signal cables and corrupting camera images.*

SECTION 11 — INTERNAL COMPONENT LAYOUT — COMPLETE MAP

11.1 Layout Principles

The internal layout follows four rules in priority order: (1) Optical components must have unobstructed clear zones to their respective windows, (2) HV wiring must be segregated from signal wiring, (3) Heaviest components must be distributed to avoid tipping the centre of gravity off-centre (to maintain level mounting), (4) Components requiring most frequent maintenance access must be nearest the tray slide-out direction (rear).

11.2 Top View Layout (738mm × 288mm internal, looking down)



11.4 Nuts, Bolts and Fasteners — Complete List

Fastener	Size	Qty	Used For	Torque
Lid perimeter bolts	M4 × 12mm, A4 SS hex-socket	32	Seal lid to body	2.5 Nm
Internal tray to body	M5 × 10mm, A4 SS hex-socket	8	Fix tray in enclosure	4 Nm
Strobe reflector to tray	M4 × 8mm, A4 SS hex-socket	4 per strobe × 3 = 12	Fix reflectors to tray	2.5 Nm
Cap bank PCB to tray	M4 × 16mm, A4 SS, with neoprene standoff	4 per bank × 3 = 12	Vibration-isolated fixing	Hand tight + 1/4 turn
HV charger PCB to tray	M3 × 6mm, A4 SS hex-socket	4 per charger × 3 = 12	Fix charger PCBs	0.5 Nm
ZED 2i to bracket	M3 × 8mm, A4 SS socket	4	Camera to aluminium bracket	0.6 Nm, with Loctite 222
ZED 2i bracket to tray	M4 × 10mm, A4 SS socket	4	Bracket to tray	2.0 Nm
Fisheye camera to bracket	1/4-20 UNC × 12mm, SS	1	Camera tripod mount	Hand tight + 1/4 turn
RGB camera to bracket	M3 × 8mm, A4 SS socket	2	Camera to bracket	0.6 Nm
LiDAR to bracket	M3 × 8mm, A4 SS socket	4	LiDAR to tray-mounted bracket	0.8 Nm
FPGA board to tray	M3 × 6mm, A4 SS, nylon standoffs 10mm	4	PCB to tray, 10mm clearance	0.3 Nm
GigE switch to tray	M3 × 6mm, A4 SS, neoprene pad	4	Switch to tray	0.4 Nm
Power distribution PCB to tray	M3 × 6mm, A4 SS, nylon standoffs 10mm	4	PCB to tray	0.3 Nm
Fan guards to fans	M3 × 8mm, A4 SS	4 per fan × 2 = 8	Guard to fan frame	0.5 Nm

Window frames (all 7)	M2.5 × 5mm, A4 SS, hex socket	4 per window × 7 = 28	Glass retaining ring to frame	0.3 Nm
HV trigger PCBs to reflector	M2 × 4mm, A2 SS	4 per PCB × 3 = 12	Trigger PCB to reflector side	0.2 Nm
Roof rack rails to vehicle	M6 × 20mm, A4 SS, with backing plate	6 per rail × 2 = 12	Rail to roof — installed once	8 Nm + Loctite 243
T-clamp bolts	M8 × 30mm, A4 SS hex	4	Box to rail clamps	12 Nm — critical torque
Anti-vibration mounts	M8 × 25mm stud, integral to mount	4	Box base to T-clamps	10 Nm
Safety cable anchor	M8 × 20mm, A4 SS eye bolt	2	Safety cable to box base	10 Nm
Nyloc nuts on T-clamps	M8 A4 SS nyloc	4	Secondary retention on T-clamps	12 Nm

SECTION 12 — AI PROCESSING PIPELINE (Roof Box Data)

All AI processing runs on the Jetson AGX Orin inside the vehicle equipment rack — not inside the roof box. The roof box is purely the sensor head. The following describes the complete real-time data flow from sensor to final RL output for both roadside signs and gantry signs.

12.1 Roadside Sign Measurement — Step by Step

Step	Location	Action	Time
1. LiDAR scans ahead	Roof box LiDAR	Point cloud processed every 100ms — searches for vertical structures at road edge 1.5–5m high	100ms cycle
2. Sign post detected at 40m	Jetson AI	Thin vertical cluster identified as sign post — distance tracking begins at 10 Hz	Continuous
3. Flash time pre-computed	Jetson + FPGA	Vehicle speed (GPS) used to predict exact timestamp when vehicle reaches 15m from sign	Updated 10×/sec
4. YOLOv10 pre-classification	Jetson — ZED 2i feed	Sign type and IRC code identified from 40m→20m approach. Correct RL threshold pre-loaded	Running continuously
5. FPGA executes flash	Roof box FPGA	At pre-computed timestamp: Strobe 1 fires (1μs, 50,000 lux). ZED 2i triggered simultaneously	T = 0

6. ZED 2i captures image	Roof box — ZED 2i	Full-resolution stereo image at exact strobe moment. Stereo depth calculated	T + 5ms
7. Distance to sign measured	Jetson GPU	ZED 2i stereo disparity → distance accurate to ±1% at 15m	T + 8ms
8. Illuminance calculated	Jetson software	E = Strobe power ÷ (4π × distance²) — corrected for strobe angle	T + 10ms
9. Sign face luminance extracted	Jetson — YOLOv10	YOLOv10 segments retroreflective zone. Mean pixel brightness → cd/m² via photometric calibration	T + 15ms
10. RL calculated	Jetson software	RL = L ÷ E in cd/lux/m². Multiple captures averaged. Outliers removed	T + 20ms
11. Angle correction applied	Jetson software	LiDAR-measured sign tilt angle used to correct for non-perpendicular illumination	T + 22ms
12. GPS tagging + compliance	Jetson + ZED-F9P	RL tagged with sign GPS position (vehicle GPS + LiDAR lateral offset). Checked vs IRC 67 threshold	T + 25ms
13. Storage + report	Jetson NVMe	RL value, sign type, IRC code, photo, GPS, Pass/Fail — written to SSD + queued for upload	Background

12.2 Gantry Sign Measurement — Step by Step

Step	Location	Action
1. Gantry detected at 100m	Roof box LiDAR + Jetson	Horizontal bar of points at 5–7m spanning road width identified as gantry portal
2. 3D structure mapping	Jetson	Post positions, crossbeam height, panel count, panel positions, panel tilt angles all extracted
3. Multi-flash sequence planned	Jetson → main FPGA	Timestamps for 2–3 flashes (at 30m, 22m, 16m ahead) pre-computed and sent to roof box FPGA
4. Panel pre-identification	Jetson — ZED 2i + RGB cam	From 50m→20m: YOLOv10 identifies each panel content, sheeting grade, MoRTH threshold pre-loaded
5. Flash 1 at 30m	Roof FPGA — Strobes 2+3	Both upward strobes fire simultaneously. ZED 2i captures. Optimised for leftmost panels
6. Flash 2 at 22m	Roof FPGA — Strobes 2+3	Second strobe pair. ZED 2i captures. Optimised for centre panels
7. Flash 3 at 16m (if needed)	Roof FPGA — Strobes 2+3	Third strobe pair for rightmost panels on wide gantries (>15m)

8. Fisheye trigger at gantry centre	Roof FPGA → Theta Z1	When vehicle directly under crossbeam: fisheye fires with upward Strobe 2+3 for overhead validation
9. Per-panel RL extraction	Jetson AI	Each panel segmented from images. Stereo depth per panel. RL = L/E with full geometric angle correction
10. Cross-validation	Jetson software	Stereo RL vs fisheye RL compared. If disagreement >20%: flag for review
11. Structural inspection flags	Jetson — LiDAR + fisheye	Bent posts, tilted panels, missing fasteners, illegal banners detected from fisheye image + point cloud
12. Per-panel reporting	Jetson NVMe + 4G	Each panel: GPS, panel ID, RL value, threshold, Pass/Fail, photo — individual maintenance order if failed

12.3 AI Models Running for Roof Box Data

AI Model	Architecture	Input	Output
Sign Detector	YOLOv10-medium (India trained, 200k+ images)	ZED 2i + RGB camera frames	Sign type, IRC code, bounding box, confidence
Sign Segmenter	YOLOv10-seg sub-model	Cropped sign image	Retroreflective zone mask (excludes borders, text, symbols)
Sheeting Classifier	MobileNetV3-small (fine-tuned)	High-res crop of sign surface	EG / HIG / DG classification from micro-texture
Gantry Panel Mapper	Custom 3D object detection on LiDAR	Point cloud segment	Panel count, 3D bounding boxes, tilt angles per panel
Angle Corrector	Physics-based geometric model	LiDAR geometry + vehicle IMU data	Entrance/observation angle → standard geometry RL correction factor
Degradation Predictor	LSTM 2-layer, 64 hidden units	Historical RL at this chainage (from HMS database)	Predicted days until RL falls below IRC 67 threshold
Anomaly Detector	Isolation Forest (sklearn)	RL readings over last 5 km section	Flags statistically anomalous sign clusters — batch defects etc.

SECTION 13 — COMPLETE COMPONENT COST BREAKDOWN

All prices in Indian Rupees (INR) as of 2024–25. International components include 15% import duty and 18% GST. Single-unit prototype pricing. Volume production (10+ units) typically 30–40% lower.

13.1 Enclosure and Mechanical

#	Component	Model / Source	Qty	Unit Cost (₹)	Total (₹)
1	Aluminium 6061-T6 enclosure (750×300×250mm, custom fab)	Local CNC fabrication, Pune/Bengaluru	1	42,000	42,000
2	AR-coated borosilicate optical windows (various sizes)	Edmund Optics / Thorlabs import	7	4,800	33,600
3	EPDM O-ring 4mm section, continuous extrusion	Parker Hannifin or local rubber supplier	1 set	2,200	2,200
4	Internal aluminium mounting tray (3mm 6061-T6)	Local fabrication, custom cut	1	7,500	7,500
5	Strobe polycarbonate safety sub-housing ×3	Local plastic fabrication	3	2,000	6,000
6	Aluminium HV divider strip (2mm, 700×60mm)	Local aluminium stock	1	800	800
7	M4 × 12mm A4 SS hex-socket bolts (lid)	Local fastener supplier	35	45	1,575
8	M5, M3, M2.5, M2 stainless fasteners (all internal)	Local fastener supplier	1 lot	3,500	3,500
9	Nyloc nuts M8, M4, M3 assortment	Local fastener supplier	1 lot	800	800
10	Neoprene rubber standoffs, pads, vibration isolators	Local rubber supplier	1 lot	2,200	2,200
11	NOA 61 UV optical adhesive + EPDM gasket material	Norland Products import via distributor	1 set	3,500	3,500
12	Black silicone RTV sealant (Dow Corning 732)	Local industrial supplier	2 tubes	600	1,200
13	Hydrophobic window coating kit	Rain-X or industrial equivalent	1	1,200	1,200
14	Assembly labour — enclosure and mechanical	Skilled technician, 20 hours	20 hrs	800/hr	16,000
TOTAL	Enclosure and Mechanical				1,22,075

13.2 Roof Mounting System

#	Component	Model / Source	Qty	Unit Cost (₹)	Total (₹)
15	Aluminium roof rack rail system (1400mm rails × 2)	Rhino Rack, Thule, or local equivalent	1 set	12,000	12,000
16	T-slot clamps (heavy duty, M8, rated 500kg each)	Bosch Rexroth or RS Components import	4	1,800	7,200
17	Anti-vibration mounts M8, 40 Shore A, 25mm dia	Trelleborg or Elesa+Ganter import	4	1,500	6,000
18	Safety retention cables 8mm SS wire × 2	Local rigging supplier	2	1,200	2,400
19	Safety carabiners + swage fittings	Local marine/rigging supplier	1 set	1,500	1,500
20	Butyl mastic sealant tape (roof rail fixing)	3M or Sika, local	1 roll	800	800
21	M6 SS bolts + aluminium backing plates (rail fix)	Local fastener + fabrication	1 lot	2,000	2,000
22	Cable conduit + grommets (roof-to-cabin routing)	Local electrical supplier	1 lot	1,800	1,800
TOTAL	Roof Mounting System				33,700

13.3 Strobe Illumination System (×3 complete units)

#	Component	Model / Source	Qty	Unit Cost (₹)	Total (₹)
23	Xenon flash tubes (Perkin-Elmer FX-4400 or equiv)	EG&G / Excelitas import via distributor	3	8,500	25,500
24	Parabolic reflector assemblies (custom machined + coated)	Local CNC fab + vacuum coating, Pune	3	18,000	54,000

25	Electrolytic capacitors 400V 100µF flash grade (Nichicon)	Mouser Electronics import	18	1,800	32,400
26	Capacitor bank PCBs (custom, 4oz copper, FR4)	PCBWay fabrication + components	3	8,500	25,500
27	High-voltage charging circuits (STM L6598 based)	Custom PCB design + fab, PCBWay	3	6,500	19,500
28	HV trigger transformer (1:800 ratio, EG&G FY-632)	Excelitas / Richardson Electronics import	3	4,200	12,600
29	HV trigger PCB (MOSFET + transformer, custom)	PCBWay fab + components	3	2,500	7,500
30	HV rated silicone wire (10kV, 0.5mm², per metre)	Belden or local HV wire supplier	3 m	800/m	2,400
31	Strobe heatsinks (Fischer SK104 or equiv, per strobe)	Fischer Elektronik import	3	2,200	6,600
32	Thermal paste (Shin-Etsu X-23-7783D)	Local electronics distributor	1 tube	1,200	1,200
33	Bleed resistors 2.2MΩ 5W wirewound	Local electronics market	3	150	450
34	HV charge indicator LEDs (red+green pairs)	Local electronics	3 sets	200	600
TOTAL	Strobe Illumination System				1,87,750

13.4 Camera System

#	Component	Model / Source	Qty	Unit Cost (₹)	Total (₹)
35	Stereolabs ZED 2i stereo camera	Stereolabs import via authorised distributor	1	1,58,000	1,58,000
36	Ricoh Theta Z1 360° fisheye camera	Ricoh India / authorised dealer	1	78,000	78,000
37	Basler acA2040-90uc colour camera	Basler AG import, authorised distributor	1	52,000	52,000
38	Kowa LM16JC 16mm C-mount lens (RGB camera)	Kowa / CBC import	1	9,500	9,500

39	Camera mounting brackets (custom aluminium, all 3)	Local CNC fabrication	3	3,500	10,500
40	USB 3.2 hub (industrial, 4-port, 12V powered)	Terminus FE8.1 or equivalent import	1	4,500	4,500
41	USB 3.0 to GigE adapter (UGREEN or Club3D, 1Gbps)	Local import e-commerce	1	3,200	3,200
TOTAL	Camera System				3,15,700

13.5 LiDAR System

#	Component	Model / Source	Qty	Unit Cost (₹)	Total (₹)
42	Livox Mid-360 LiDAR	Livox Technology (DJI subsidiary) import	1	1,04,000	1,04,000
43	LiDAR internal mounting bracket (custom aluminium)	Local CNC fabrication	1	2,200	2,200
TOTAL	LiDAR System				1,06,200

13.6 FPGA, Power, and Networking Electronics

#	Component	Model / Source	Qty	Unit Cost (₹)	Total (₹)
44	Local FPGA board (Digilent Arty A7-100T)	Digilent import	1	11,500	11,500
45	DC-DC isolated converter TRACO TEN 60-1212WI	TRACO Power import via Mouser	1	7,800	7,800
46	5V DC-DC switcher Murata OKI-78SR-5	Murata import via Mouser	2	650	1,300
47	±5V linear regulators (TI TPS7A4901/7901 pair)	TI distributor (Arrow/Mouser)	1 pair	1,200	1,200
48	Power distribution PCB (custom, heavy copper)	PCBWay 4oz copper fabrication + components	1	6,500	6,500
49	Phoenix Contact MKDS terminal blocks (5-pin)	Phoenix Contact distributor	3	800	2,400
50	EMI filter (Schurter FMAB, 10A)	Schurter import	1	2,200	2,200

51	Internal GigE switch 5-port (TP-Link TL-SG1005, modified)	TP-Link local + modification labour	1	2,500	2,500
52	Signal conditioning + status PCB (PIC12F1840 based)	Custom PCB, PCBWay + components	1	4,500	4,500
53	Fuse holders ATO + assorted mini-blade fuses	Local auto electronics market	1 lot	800	800
TOTAL	FPGA, Power, Networking Electronics				40,700

13.7 Thermal and Environmental Management

#	Component	Model / Source	Qty	Unit Cost (₹)	Total (₹)
54	PTC heater strips 40W each, 12V self-regulating	Aliexpress industrial / local	2	1,400	2,800
55	Delta BFB0812HH cooling fans 80mm, 12V, 4500RPM	Delta Electronics import	2	1,800	3,600
56	Fan wire guards 80mm stainless	Local electronics market	2	400	800
57	DS18B20 temperature sensors (waterproof, cabled)	Local electronics market	3	200	600
58	Sensirion SHT31 humidity sensor module	Sensirion import / Mouser	1	2,000	2,000
59	Silica gel desiccant 100g cartridges with indicator	Local chemical/industrial supplier	2	500	1,000
60	Aluminium fin heat exchange plate 700×200×3mm	Local aluminium fabrication	1	3,500	3,500
TOTAL	Thermal and Environmental				14,300

13.8 External Connectors and Wiring

#	Component	Model / Source	Qty	Unit Cost (₹)	Total (₹)
61	Amphenol C016 4-pin waterproof connector pair (power)	Amphenol import via distributor	1 pair	3,200	3,200
62	LEMO ERA 8-pin circular connector pair (data)	LEMO import via distributor	1 pair	8,500	8,500
63	Binder 723 6-pin circular connector pair (timing)	Binder import via distributor	1 pair	4,200	4,200

64	Internal wiring harness (all cables, labelled, assembled)	Cables, connectors, labels, trunking, labour	1 lot	12,000	12,000
65	External cables — roof to vehicle (5m, all 3)	Belden shielded cable + connectors, local assembly	1 set	6,500	6,500
66	Split-loom conduit 25mm + P-clips (external routing)	Local electrical supplier	1 lot	1,500	1,500
67	Roof cable grommet (rubber, adhesive-backed)	Local rubber supplier	1	600	600
TOTAL	Connectors and Wiring				36,500

13.9 Assembly, Testing and Miscellaneous

#	Component	Model / Source	Qty	Unit Cost (₹)	Total (₹)
68	Roof rack installation on vehicle (one-time)	Workshop labour, 3 hours	3 hrs	1,000/hr	3,000
69	System assembly labour (mechanical + electrical)	Skilled technician, 60 hours	60 hrs	800/hr	48,000
70	High-voltage safety testing + compliance check	Lab time + equipment	1 lot	15,000	15,000
71	Photometric calibration — strobe output measurement	Calibrated luxmeter + lab session	1 lot	12,000	12,000
72	Camera photometric calibration (luminance response)	Calibrated luminance source + lab session	1 lot	18,000	18,000
73	System integration test — full sign + gantry test	On-road test session, dummy signs	1 lot	20,000	20,000
74	Spare parts kit (flash tubes ×2, capacitors ×6, fuses)	First-year maintenance kit	1 lot	18,000	18,000
75	Safety stickers, labels, serial plate	Local printing + stainless nameplate	1 lot	2,000	2,000
TOTAL	Assembly, Testing, Miscellaneous				1,36,000

SECTION 14 — GRAND TOTAL COST SUMMARY

#	Category	Items	Qty Items	Subtotal (₹)	% of total
1	Enclosure and Mechanical	14 items	14	1,22,075	13.3%
2	Roof Mounting System	8 items	8	33,700	3.7%
3	Strobe Illumination System (×3)	12 items	36	1,87,750	20.4%
4	Camera System	7 items	7	3,15,700	34.3%
5	LiDAR System	2 items	2	1,06,200	11.5%
6	FPGA, Power, Networking	10 items	10	40,700	4.4%
7	Thermal and Environmental	7 items	7	14,300	1.6%
8	Connectors and Wiring	7 items	7	36,500	4.0%
9	Assembly, Testing, Misc	8 items	8	1,36,000	14.8%
TOTAL	ROOF UNIT — COMPLETE BUILD COST	75 items total		9,19,925	100%
	Rounded estimate (prototype unit)			~Rs. 9.2 Lakh	(~$11,000 USD)

NOTE: *Volume production (10+ units): Estimated 35–40% reduction = approximately Rs. 5.5–6.0 lakh per unit. The ZED 2i stereo camera (Rs. 1.58 lakh) and Livox LiDAR (Rs. 1.04 lakh) together account for 28% of total cost. The three strobe assemblies with capacitor banks account for another 20%. These six items drive 48% of total cost and are where volume sourcing efforts should be focused.*

END OF ROOF-MOUNTED BOX TECHNICAL REFERENCE